

Manifold for T1D – Restoration

β -cell preservation, regeneration, and replacement-therapy analytics

Type 1 Diabetes does not end at diagnosis — three clinical paradigms are now converging on the restoration of β -cell function: **preservation** (teplizumab post-diagnosis, verapamil, ATG, GLP-1 combinations), **regeneration** (DYRK1A inhibitors, senolytics), and **replacement** (Vertex VX-880, Sernova Cell Pouch, ViaCyte/CRISPR). Each paradigm has actionable agents and rich longitudinal data; the analytic gap lies in matching patient to therapy and in ordering combination regimens on a principled causal graph. **Manifold** configures three engine capabilities to the domain — causal discovery on multi-omic and clinical-trial data, criticality estimators on post-intervention trajectories (mixed-effects C-peptide models, heterogeneous-treatment-effect learners, Gaussian process regression, Bayesian change-point), and Pearl do-calculus with interdiction for principled combination-therapy design.

Vertex VX-880's initial cohort achieving insulin-independence constitutes a proof-of-principle rather than a finished therapy. The outstanding question is no longer whether β -cell function can be restored, but for which patients, with which adjuncts, and over what duration — a problem of causal inference.

Ω F PILLAR MAPPING – RESTORATION PHYSIOLOGY

Ω F PILLAR	RESTORATION READING
I — Interaction	Drug $\times$ genotype $\times$ autoantibody-profile interactions for combo response; graft $\times$ host-immune compatibility
R — Reserve	Residual / post-infusion β -cell mass (C-peptide AUC on MMTT); stress-test response
J — Junction	Immune-metabolic coupling: cytokine-panel $\leftrightarrow$ C-peptide transfer entropy; autoimmunity $\times$ graft survival
C — Cascade	β -cell senescence and ER-stress cascades; DYRK1A $\rightarrow$ cell-cycle progression; inflammatory escalation post-graft
T — Temporal	Hazard of honeymoon-end, graft-failure, or response-loss given current trajectory

PROPOSED PARTNERSHIP

Partnership with a preservation-trial, β -cell-regeneration, or islet-transplant principal investigator (or equivalent cohort steward — Joslin, Mount Sinai Stewart Lab, a Vertex trial collaborator, or a HIRN consortium member) is sought for:

1. Retrospective trial or registry data access under the partnering institution's existing Data Use Agreement.
2. Clinical co-authorship on the heterogeneous-treatment-effect / graft-survival benchmark manuscript (twelve to fourteen weeks from data access).
3. Co-application to Breakthrough T1D Cures, Helmsley Restoring Insulin Production, or NIH NIDDK HIRN for the Phase 2 prospective arm.

Apex Analytica contributes the causal-inference engine, the Ω F framework, and the engineering team. Full brief: M-T1D-02 .